MACHINE LEARNING & BUSINESS APPLICATIONS

Yeti: A Churning Case Study

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The Problem

- Yeti Travel Agency specializes exclusively in **school trip packages**
- Returning customers are the backbone of the revenue, but recently detected a **drop in sales among regular customers**
- Data exists but is **not centralized** → split across departments, incomplete and inconsistent

~40%

Churn Rate

The Goal

1. Predictive Model

Build a model to predict churn probability
Retained = 1 (school came back) or
Retained = 0 (school churned)
→ **Binary Classification** problem

2. Feature Analysis

Identify the most **informative features** to design specific strategies to strengthen customer loyalty and reduce churn risk

3. Recommendations

Use the insights from the model to design specific strategies to **strengthen customer loyalty** and **reduce churn risk**

The Pipeline

01
Data & Preprocessing

02
Data Exploration

03
Comparing Models

04
Winner(s) & Results

05
Recomm-endations

01 Data & Preprocessing

Data

Sales
- 4153 instances
- 24 fields

Financial
- 4151 instances
- 12 fields

Customer Relationship Mgmt
- 4148 instances
- 20 fields

Preprocessing

01 ***ID Normalisation***

Each dataset had its own ID format (e.g. CC1387A, 1387, 01387I). Extracted the inner number as the key.

02 ***Join***

Joined on the new key.
3095 rows retained.

03 ***Cleaning and Formatting***

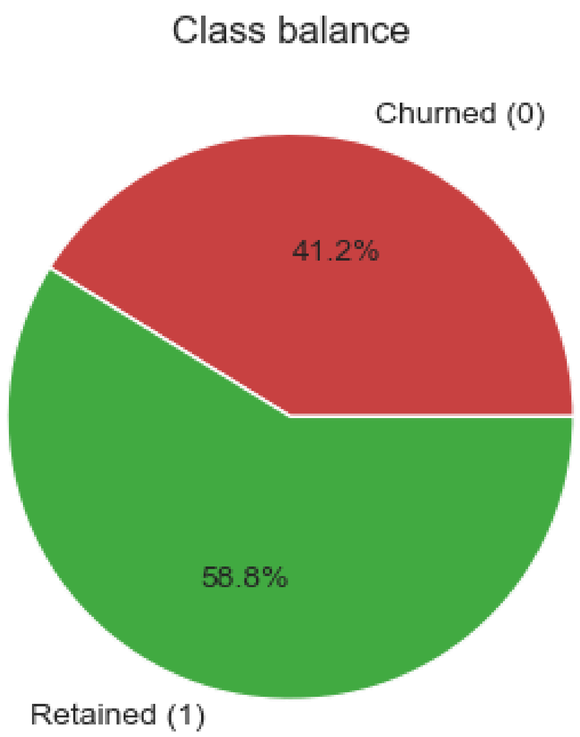
Fixed decimal conversion and created new date columns. 75/25 split.

02 Exploratory Data Analysis

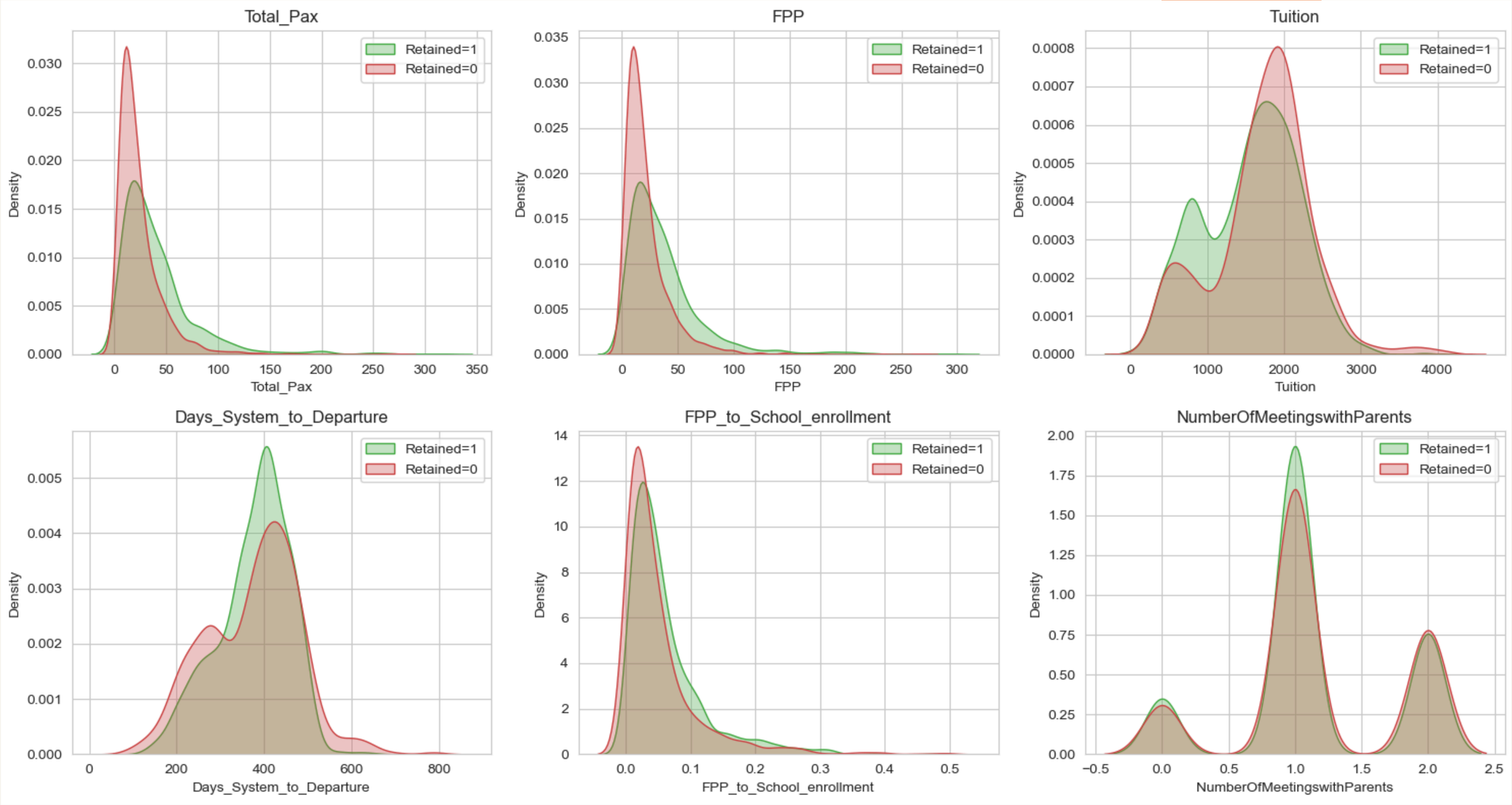
Class Balance

Class Imbalance

- 3,095 training records
- 58.8% retained and 41.2% churned
- Slightly imbalanced does not require resampling techniques



Key Features



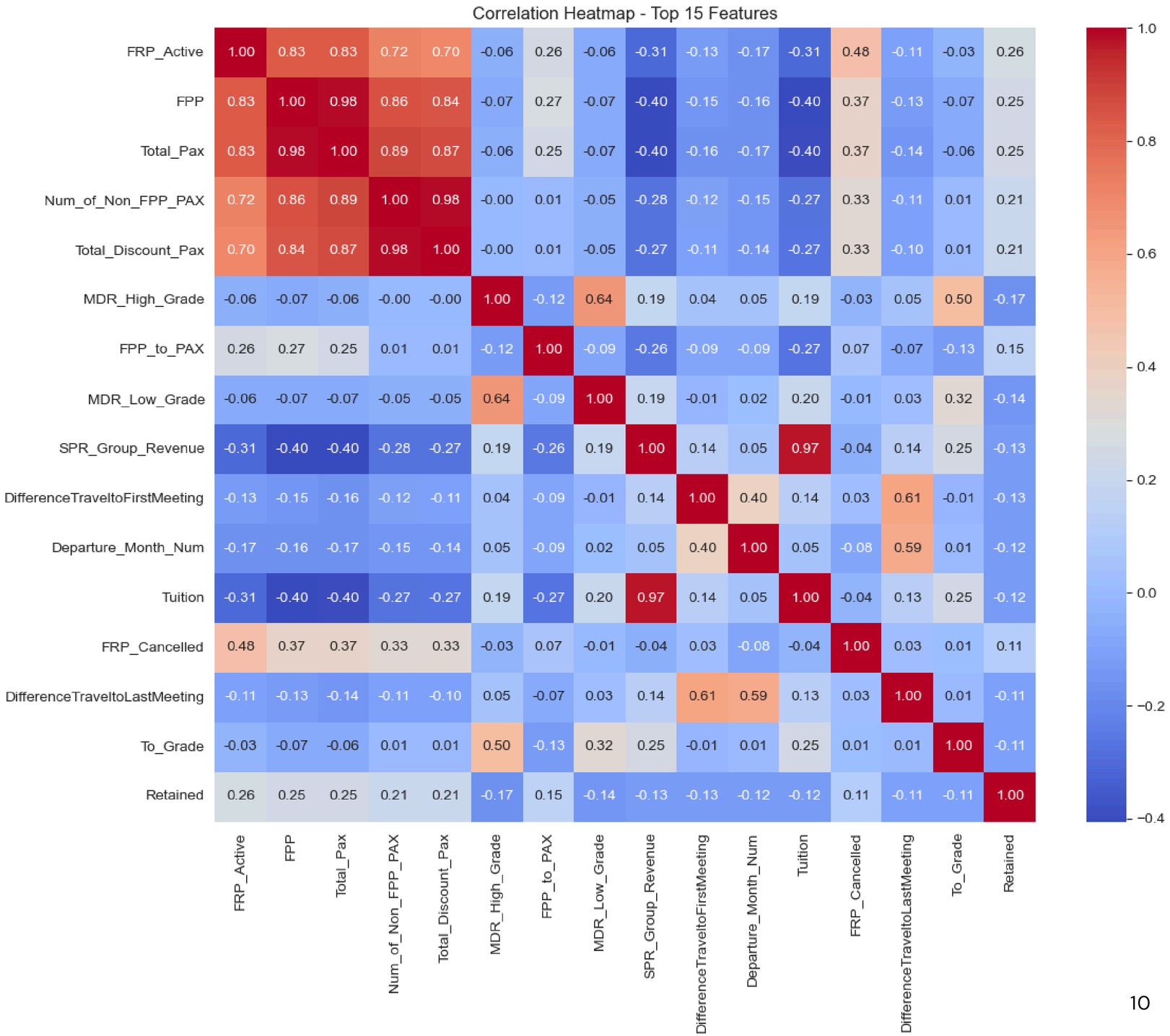
What separates retainers from churners

- **Planning early.** Retained schools plan ~400 days ahead. Churners ~200.
- **Group size.** Churn concentrates in smaller groups.
- **Parent meetings.** More meetings → higher retention.

Feature Correlations

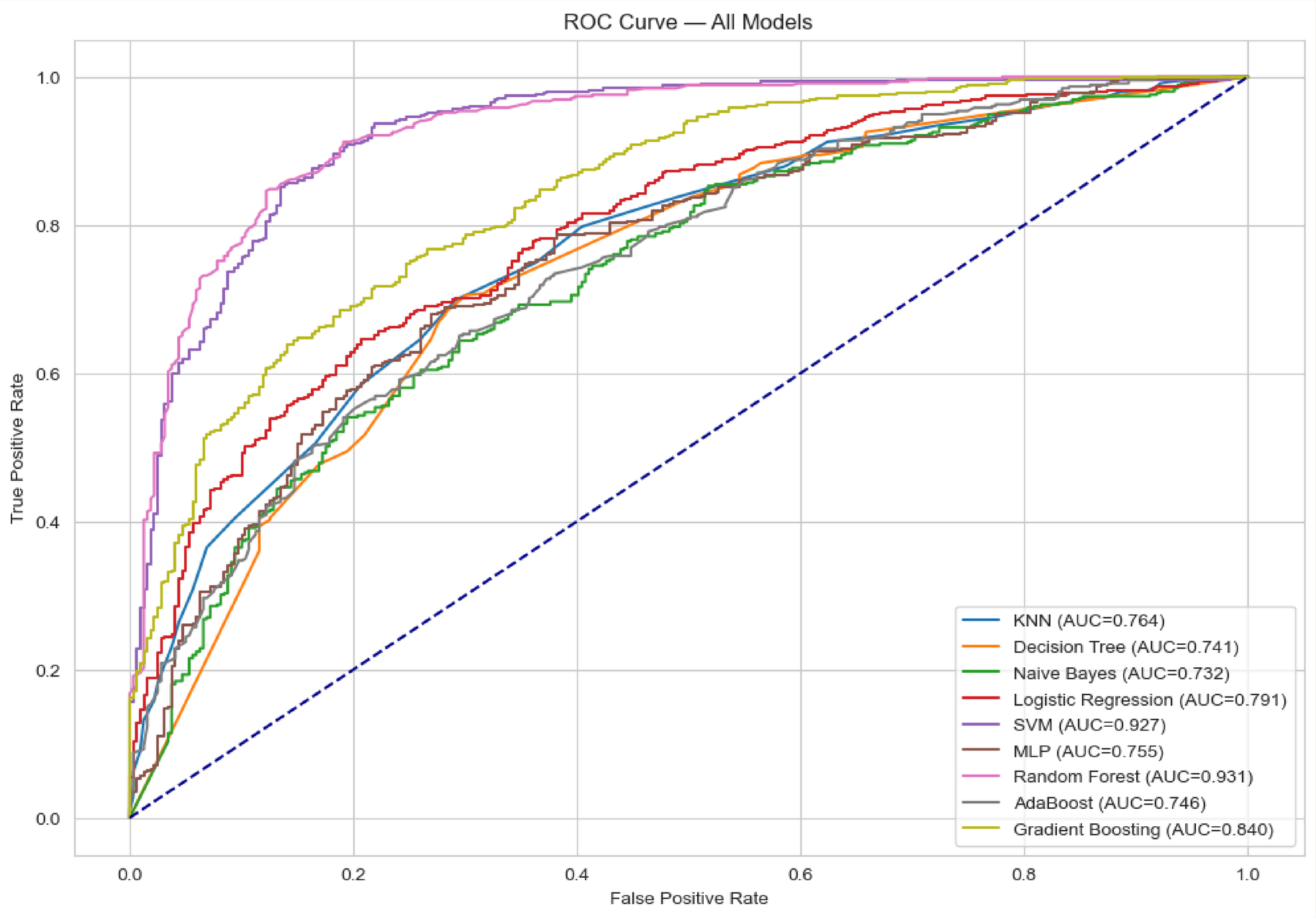
Strongest Predictors

- ***FRP_Active, FPP, Total_Pax***: large, financially committed groups return.
- Schools with larger groups and more full-payment participants are more likely to return
- Active cancellation insurance (FRP) signals trust in the agency → higher retention
- Late-planning and lower school enrolment features pull toward churn.

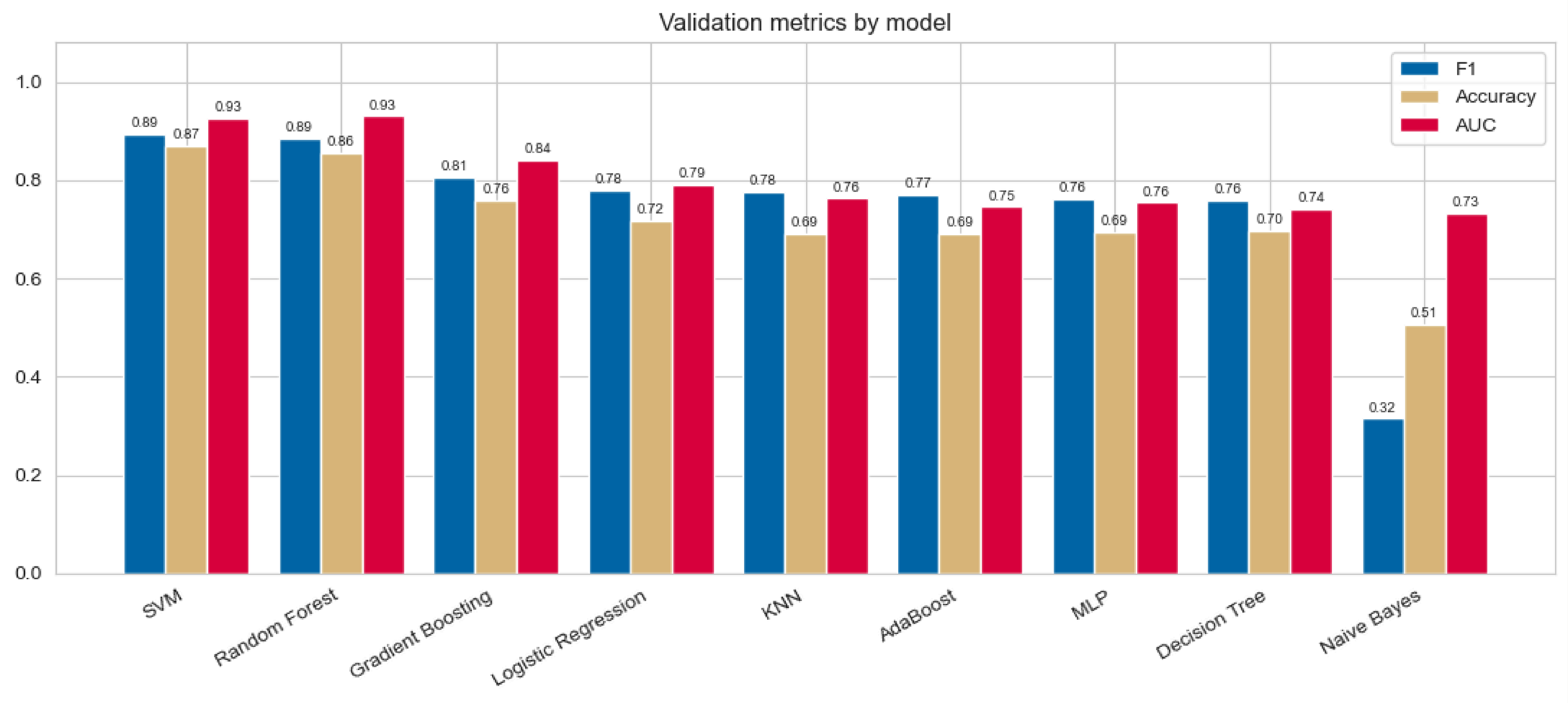


03 Model Comparison

ROC Curve


ROC Curve — All Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
KNN (AUC=0.764)
Decision Tree (AUC=0.741)
Naive Bayes (AUC=0.732)
Logistic Regression (AUC=0.791)
SVM (AUC=0.927)
MLP (AUC=0.755)
Random Forest (AUC=0.931)
AdaBoost (AUC=0.746)
Gradient Boosting (AUC=0.840)

Metrics


Validation metrics by model
F1
Accuracy
AUC
1.0
0.8
0.6
0.4
0.2
0.0
0.89
0.87
0.93
0.89
0.86
0.93
0.81
0.76
0.84
0.78
0.72
0.79
0.78
0.69
0.76
0.77
0.69
0.75
0.76
0.69
0.76
0.76
0.70
0.74
0.32
0.51
0.73
SVM
Random Forest
Gradient Boosting
Logistic Regression
KNN
AdaBoost
MLP
Decision Tree
Naive Bayes

04 Model Selection & Results

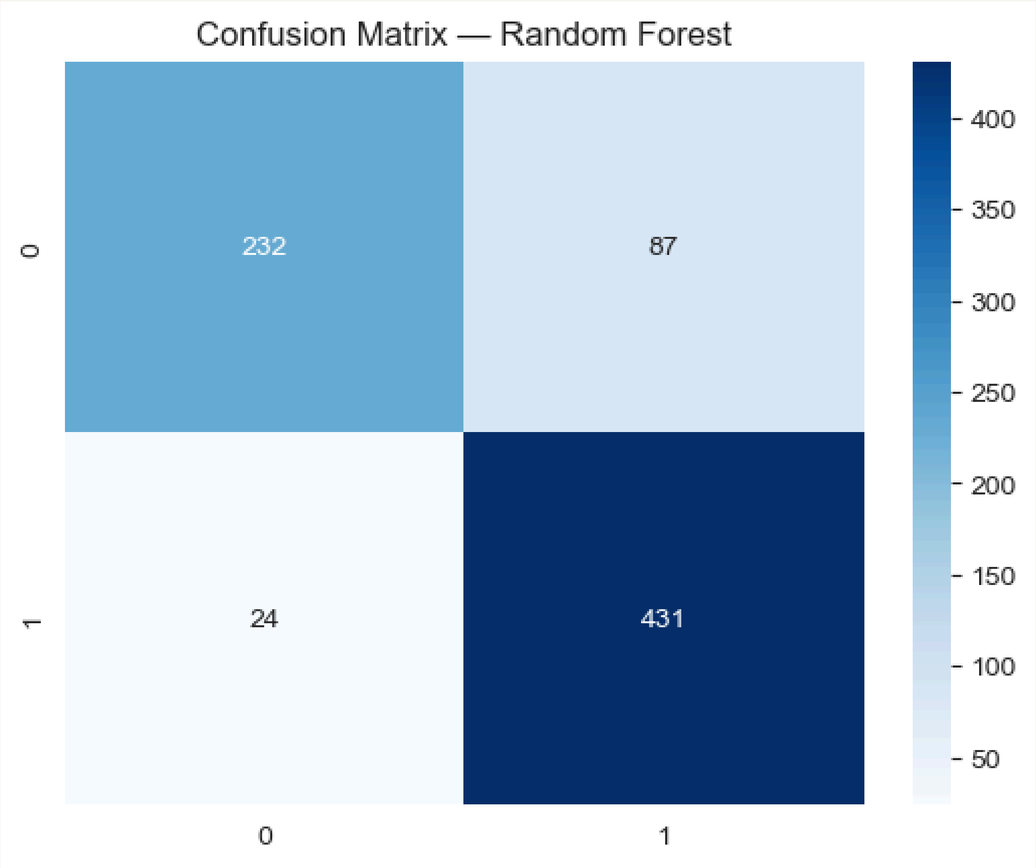
Models & Performances

Model	F1	Accuracy	AUC
SVM	**0.894**	**87%**	0.927
Random Forest	0.886	86%	**0.931**
kNN	0.790	73%	0.820
Gradient Boosting	0.807	76%	0.840
Logistic Regression	0.779	72%	0.791
AdaBoost	0.772	69%	0.746
MLP	0.763	69%	0.755
Decision Tree	0.760	70%	0.732
Naive Bayes	–	–	invalid

Random Forest

F1	0.886
AUC	0.931
ACCURACY	86%

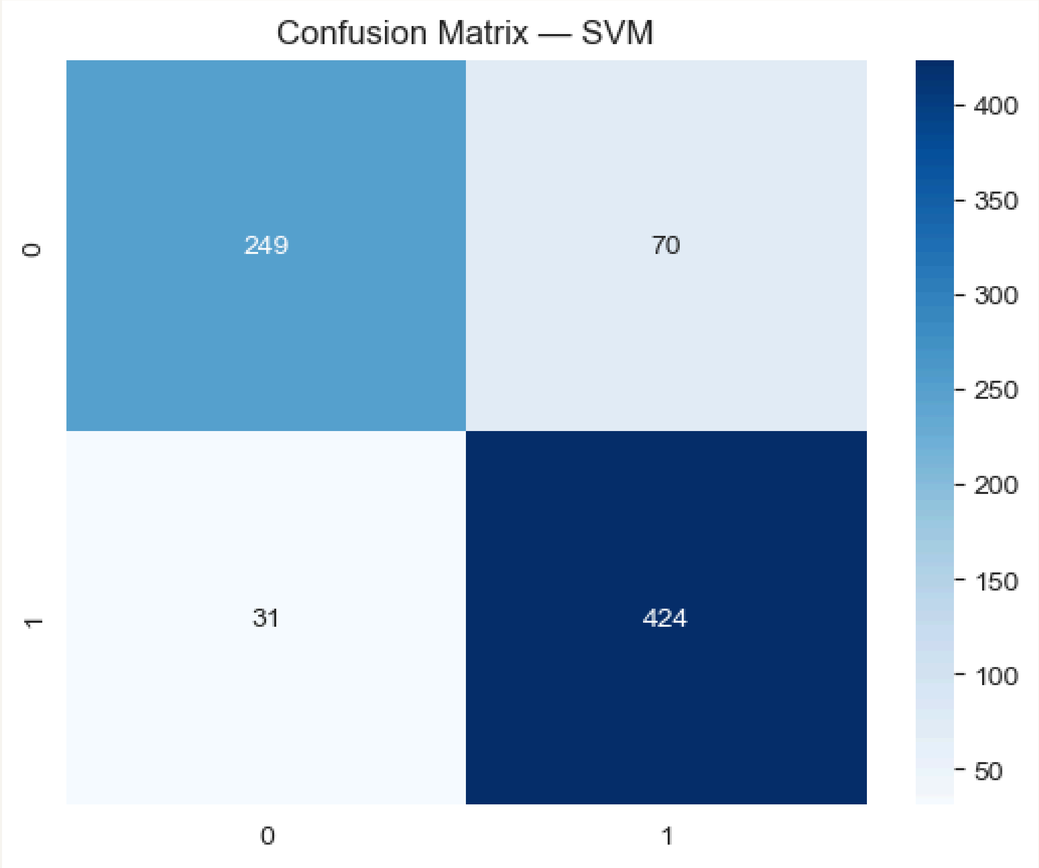
Random Forest produced similar results as SVM. With a slightly higher AUC. Also a strong and interpretable alternative.



SVM – The Chosen One

F1	0.894	▲0.008
AUC	0.927	▼0.004
ACCURACY	87%	▲~1%

SVM with RBF kernel achieved the **best balance** of F1 and accuracy across all models. The confusion matrix shows strong performance on both classes.



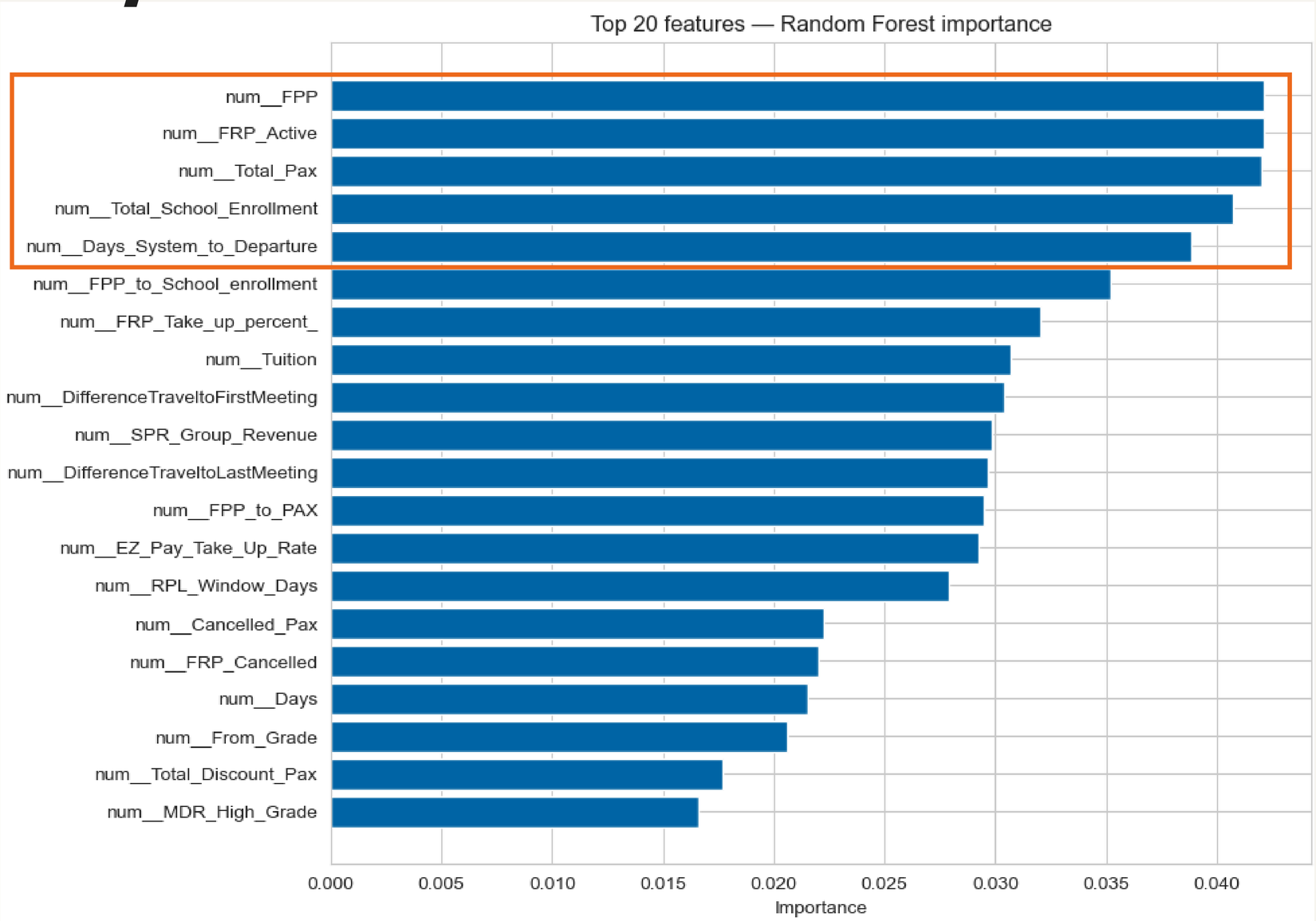
Prediction Results

630 Records

	SVM	Random Forest
Retained	396	435
	62.9%	69.0%
Churned	234	195
	37.1%	31.0%

05 Final Recommendations

Feature Importance: Loyal Schools vs. At-Risk Schools



Top 5 drivers of retention

1. ***FPP*** *(0.042)* — full paying participants return
2. ***FRP_Active*** *(0.042)* — buying insurance signals commitment
3. ***Total_Pax*** *(0.042)* — large groups of passengers return
4. ***Total_School_Enrollment*** (0.041) — bigger schools → bigger groups → bigger trips
5. ***Days_System_to_Departure*** *(0.039)* — early planners stay

Characteristics & Recommendations for Loyal Schools

Loyal	Retention Strategies
Large group	👥 Referral program
Big school	🎯 Personalized packages
Plans early	🌟 Early access
Buys insurance	🏆 Loyalty program
High participation rate	🤝 VIP relationship

Characteristics & Recommendations for At-Risk Schools

At-Risk	Intervention Strategies
Small group	👥 Group growth rewards
Smaller school	🏫 Targeted packages
Books late	📅 Early booking incentives
Skips insurance	🛡️ Make Insurance default
Low participation rate	🤝 Proactive outreach

Plan for Next Quarter at Yeti

Target Bigger Sized Schools

Total enrolment and school size are predictors of business. That could also increase the number of full paying participants.

Focus on Early Outreach

Start contacting schools soon (<9mo). Early engagement was found important at retaining business.

Bundle Insurance into the Total

Schools buying insurance commit more deeply to the trip and are more likely to return. Insurance is a key driving factor.

Target Schools with Low Participation

When participation is low, churn risk increases problematically. Offer incentives to grow group size within existing bookings.

Limitations

01 ***Single-Year Snapshot***

The events analysed only span for a small period of time. For supervised learning more historical instances power better results

02 ***Correlations Only***

The project simply looks at correlation, further studies can determine causation further

03 ***Missing Data Columns***

Customer satisfaction data would be optimal for studies like this

04 ***Heavy multicollinearity***

Ranking is not robust due to multicollinearity

What's Next

01 ***Gather More Historical Data***

Re-propose the same analysis in a few years' time in order to get results more aligned to reality

02 ***Pilot a Retention Action***

A/B test one recommendation on a matched subset of customers.

03 ***Implement Satisfaction Tracking***

Proposing short experience surveys to the customers, in order to gauge satisfaction information

04 ***Eliminate noise***

Cluster correlated features into a single composite

Thank You